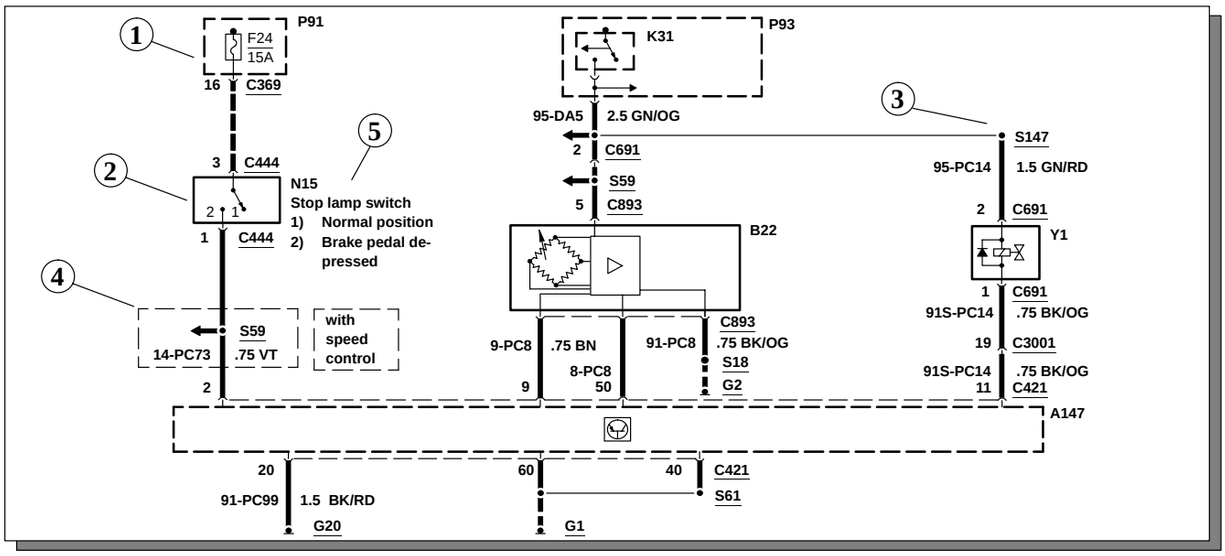
Note

All wiring connections between components are shown exactly as they exist in the vehicles. It is important to realise, however, that no attempt has been made on the schematic to represent components and wiring as they physically appear on the vehicle. For example, a 4-foot length of wire is treated no differently in a schematic from one which is only a few inches long. Furthermore, to aid in understanding electrical (electronic) operation, wiring inside complicated components has been simplified.



Complete Circuit Operation

Each circuit is shown completely and independently in one chapter or “cell”. Other components which are connected to the circuit may not be shown unless they influence the circuit operation.

Current Flow (1)

Each cell normally starts with the component that powers the circuit such as a fuse or the ignition switch. Current flow is shown from the power source at the top of the page to ground at the bottom of the page.

Switch Positions (2)

Within the schematic, all switches, sensors and relays are shown “at rest” (as if the ignition switch were OFF).

Splices (3)

An arrow indicates that the splice is not shown completely. The page where the splices appears in full is listed in the Index.

Boxes (4)

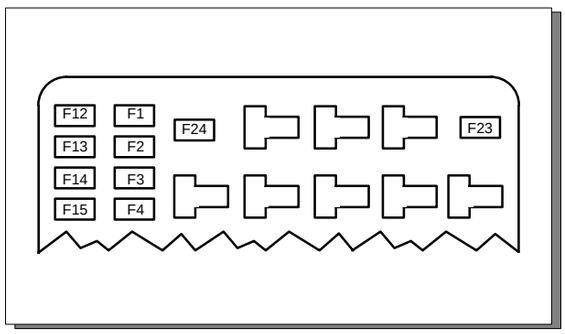
A thin dashed box on a schematic indicates a part of the circuit which is only present for a particular vehicle model, country, or option. These qualifiers will be shown next to the box on the schematic.

Component Names and Notes (5)

Component names are placed on the right hand side of each component. Any notes that describe switch positions or operating conditions follow the name. Descriptions of the internals of the component (e.g. “Speed sensor”) are also included here.

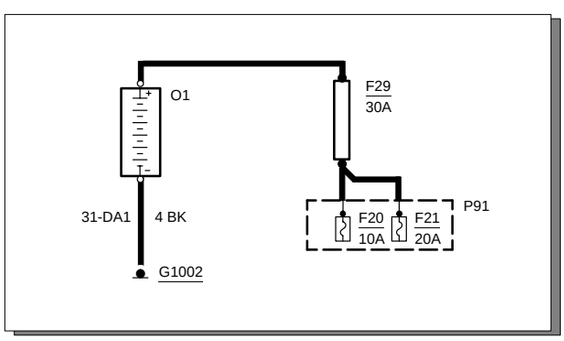
Fuse and Relay Information

The “Fuse and Relay Information” cell (700-01-00) contains a view of the Fuse-/Relay Box in which all fuses and relays are identified. Cell 700-01-00 also specifies in table format which circuits/systems are protected by each fuse.



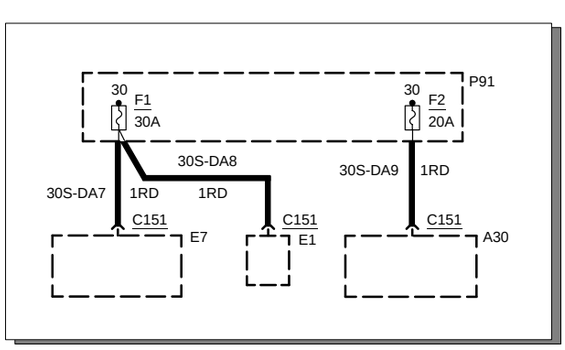
Power Distribution

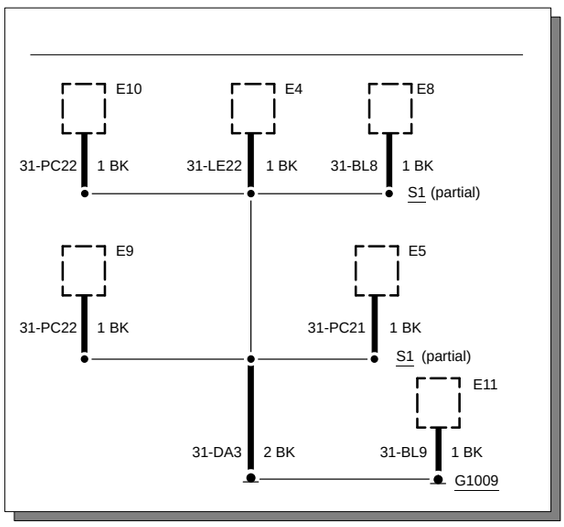
The “Power Distribution” cell (700-02-00) shows the current feed circuit. The current path is shown from the battery to the ignition switch and to all fuses.



Fuse Details

The “Fuse Details” cell (700-03-00) shows the circuits protected by each fuse. The circuit is traced from the fuse to the component. All details (wires, splices, connectors) between the fuse and the first component are shown.

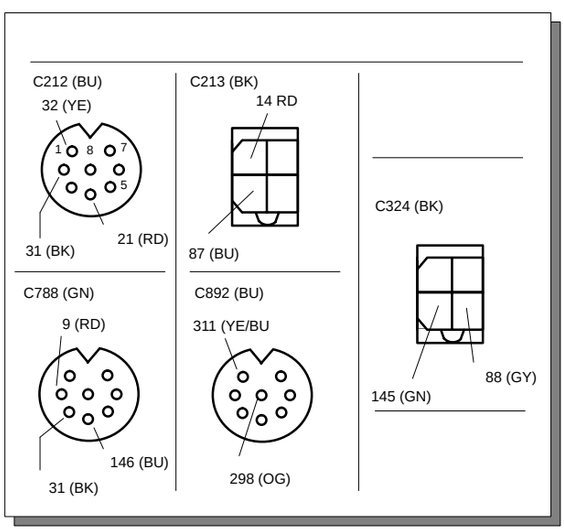




Ground Distribution

Cell 700-04-00 contains the schematics that show the complete details for each ground connection or main ground splice. This is useful in diagnosing a problem affecting several components at once (poor ground connection or ground splice). All Details (wires, splices, connectors) between the ground point and the components are shown. These ground connection details are shown here in order to keep the individual cell schematics as uncluttered as possible.

Splices with a large number of wires may be split in order to improve the graphic representation. A thin line is then used to indicate the unity of the split splice.



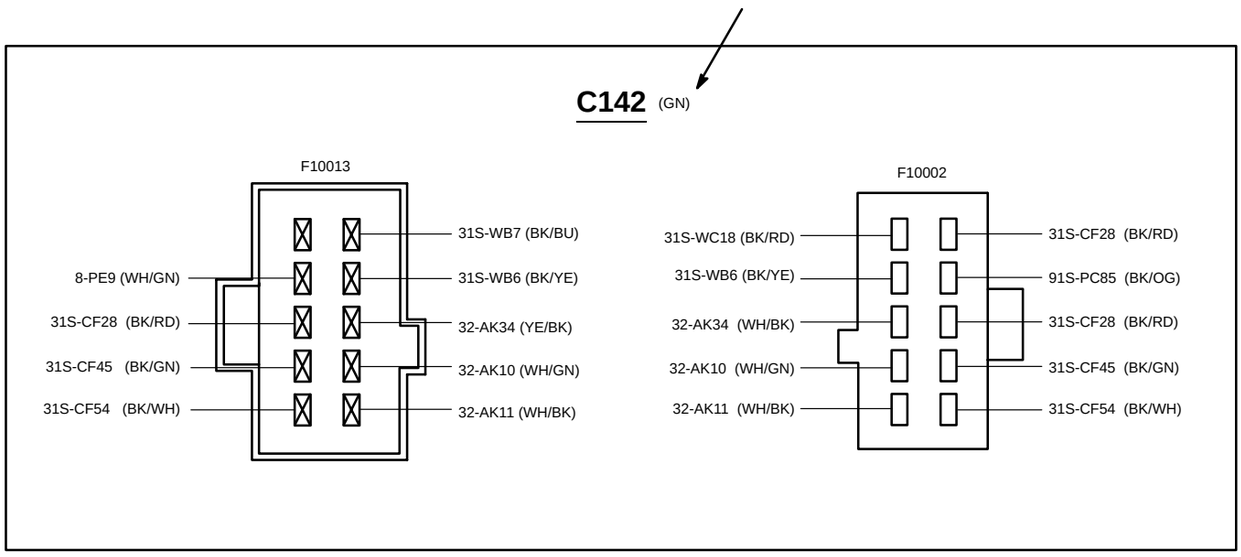
Component Identification Information

Component Identification Information begins in cell 700-06-00 and consists of
(1) Component Location Chart (List),
(2) Component Location Views, and
(3) Connector Views (Faces).
The **Component Location Chart** helps the user find where the various items depicted on the schematic can physically be found on the vehicle. A brief written description of the location is given, along with a reference to the component location views.
The **Component Location Views** show the components and their connecting wires as they can be found on the vehicle.
The **Connector Views** show the cavity or pin locations of all connectors. They are shown as they would appear to a user probing the wires with the connector disconnected.

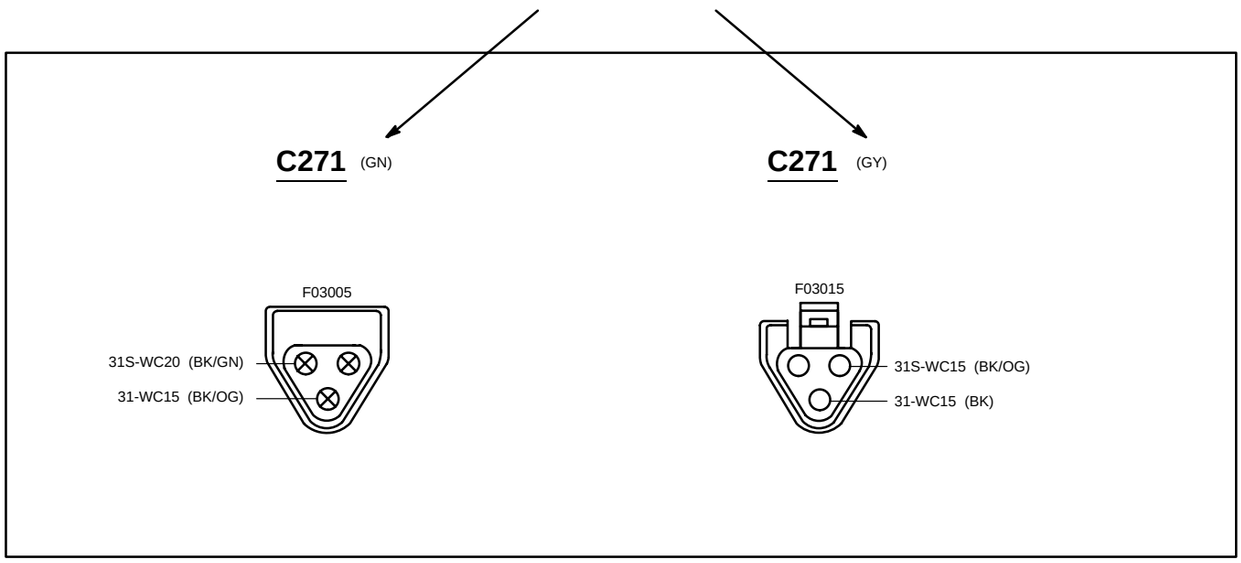
Note:

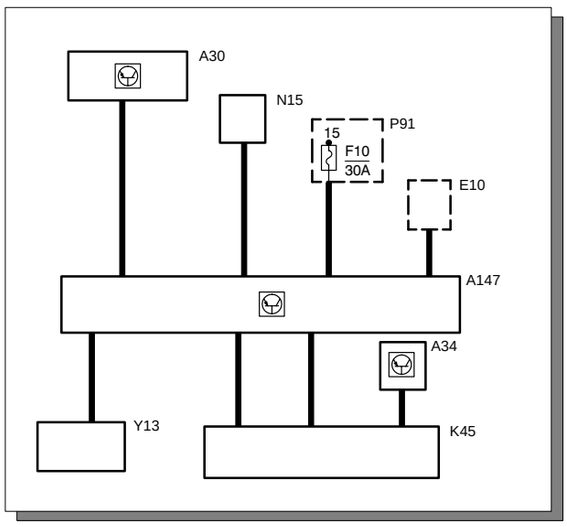
Connector colours are portrayed in the following manner:

when both sides of the connector have the same colour



when the male and female sides of the connector have different colours





Block Diagrams

Block diagrams can be found at the beginning of some cells. These diagrams provide an overview of the systems detailed on the pages that follow them. All components connected to the control module are shown in order to provide easy understanding of the system component interactions.

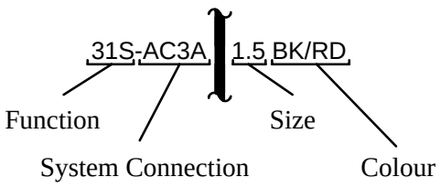
Circuit Numbering and Wire Identification

Circuit Numbering:

Ford has introduced a world-wide uniform system for circuit numbering and wire identification. The system is called Function - System Connection, or FSC for short.

FSC was developed mainly to assist in vehicle development and production processes, but is also helpful in helping the technician troubleshoot electrical circuits.

Example:



Function:

The first two digits identify the function of the wire. In this case, the function code includes the letter “S” to indicate it is an additionally switched function. The function part of the circuit number is the most helpful to the technician in troubleshooting a circuit. The function codes are based on some DIN designations used previously, with new numbers added for overload-protected functions, sensor functions, data links and special electronic module functions. A complete list of function codes and their descriptions is included in cell 100-00-04 “Symbols”.

System Connection (including branch):

Systems are related to subsets of the vehicle circuitry. A complete list of the system codes can be found in cell 100-00-04 “Symbols”.

Immediately following the system letter code is the connection number specific to that system. A branch identification is used to differentiate wires in one connection with the same function.

Wire Identification (wire colours):

The wire identification consists of a basic colour and an identification colour, and is determined directly from the wire's circuit number. In the wiring diagrams the wire colours will be located separately next to the wires as before.

The basic and identification colours are abbreviated using the international norm IEC 757. The abbreviations are listed in cell 100-00-04 "Symbols".

Each function code has a specific basic colour associated with it which can also be seen in the chart of function codes in cell 100-00-04 "Symbols". The identification colour is a coloured stripe used to differentiate several wires with the same function within one component connector.

One essential difference from the previous wire colouring system is the change of some of the basic wire colours. Ground circuits, previously assigned the basic colour brown (BN) are now black (BK).

Using the circuit number and wire identification used in the above example we know the following information about the wire.

Function: 31 = Ground
S = Additionally switched circuit

System: AC = Headlamp levelling

Connection: 3 = Switch connection
A = Branch

Size: 1.5 = 1.5 mm^2

Colour: BK = Basic colour black
(determined by function 31)
RD = Identification colour red
(red stripe)